

FiberNet Pakistan

OBJECTIVE:

Analyze customer churn and early life churn (ELC) patterns in **FiberNet Pakistan**, an optical fiber broadband ISP, using Power BI to uncover actionable insights that can help improve retention.

BACKGROUND:

FiberNet Pakistan is a growing fiber-optic ISP serving Tier 1 (Karachi, Lahore, Islamabad) and Tier 2 (Peshawar, Quetta, Multan, Faisalabad). They offer high-speed internet plans to households and small businesses, with a growing subscriber base of 200,000+ connections.

Despite strong sales, the company is facing a serious churn problem, particularly within the first 180 days of service activation.

While FiberNet is known for its blazing speeds and competitive prices, the management is concerned about rising customer churn, especially within the first six months of new connections. You are working as a Junior Data Analyst in the Customer Insights Team.

As part of the data team, you've been given access to the company's customer database. Your job is to analyze patterns in churn and early-life churn (ELC), identify high-risk customer segments and provide data-driven recommendations to help the marketing and customer support teams improve retention.

You will use your Analysis & Reporting skills to:

- Clean and query raw customer, usage, and payment data
- Investigate churn behavior across cities and acquisition channels
- Discover early signals of churn (low usage, missed payments, Non Verification of Customers)
- Present actionable insights with supporting metrics

This case study will simulate the kind of work expected in real-world roles such as:

- Data Analyst (Telecom/ISP Industry)
- Customer Insights Associate
- Business Intelligence Junior Analyst

You are expected to think critically and draw clear, evidence-based conclusions. Just like you would in a real job.

ANALYSIS:

- 1. What are the monthly churn trends over time?**
Analyze customer churn over the last 6 months.
- 2. How is churn broken down by different Categories?**
Compare Early Churn vs Core Churn and understand churn reasons such as OCR (on customer request) or CMP (due to non-payment).
- 3. Which Sales acquisition type has the highest churn?**
Evaluate churn across different types of customer acquisition: New Sales, Migration In, and Restorations.
- 4. What is the churn distribution by data usage?**
Analyze churn across different usage levels, if low or high data usage customers are more likely to churn.
- 5. Which Product type has the highest churn?**
Compare churn between customers using Copper and Fiber connections
- 6. Where is churn geographically concentrated?**
Identify zones, regions and exchanges with the highest churn.
- 7. When do Early Churn customers typically churn relative to their joining month?**
Create a churn timing matrix using Sale_Date (join date) and Year_Month (churn date) for customers labeled as Early Churn in the Churn_Flag column
- 8. Which top 2 salespersons are responsible for the most churn in each Region, Zone, or Exchange?**
Filter and group churn by geography and identify the top churn driving salespeople.
- 9. What is the single biggest root cause of churn based on the data?**
Use all available data (Churn categories, Product, Sales acquisition cost, MRC, Usage (in Gbps), KYC etc.) to determine the most significant factor driving customer churn.

Dataset Explanation

There are three tables given in CSV format. Import them into your MySQL workbench using the Table Import Wizard.

Column Name	Description
Year_Month	Month when the customer churned
Usage_GB	Internet data used by the customer before churn (in GB)
Sale_Date	Date when the customer joined the company
Billing_Account_Id	Unique customer identifier
Churn_Flag	Type of churn: Early (Acquired during previous 6 months) or Core (Other Than Early)
Customer_Name	Name of the customer
Product	Type of connection: Copper or Fiber
Sales_acquisition_type	How the customer joined: New Sales, Migration In, or Restoration
Churn_Sub_Category	Reason for churn: OCR (on request) or CMP (non-payment)
sales_created_by	Salesperson who acquired the customer
Data_Rate	Internet speed (e.g., 10 Mbps)
MRC	Monthly Price
Zone	Higher-level geographic area covering multiple regions
Region_Name	Geographic region where the customer was served
Exchange_Name	Local exchange that provided service to the customer
KYC	Whether customer identity verification was done before Sales (Yes/No)
Upfront_Amt	Whether Customer submitted upfront Device payment or not